Chapter 4
Limits and Continuity

Note 2

Lecturer: Dr. Rajitha Ranasinghe Academic Year: 2024/2025

4.2 : One-sided Limits and Limits at Infinity

In the preceding section we defined the limit of a real-valued function. In this section we expand that definition to handle more general situations.

What is the limit of $f(x) := \sqrt{x-1}$ as $x \to 1$? A reasonable answer is that the limit is zero, This function, however, does not satisfy **Definition 4.1.1** because it is not defined on an OPEN interval containing $a = 1$. Indeed, f is only defined for $x \geq 1$. To handle such situations, we introduce "one-sided" limits.

Definition 4.2.1 : Let $a \in \mathbb{R}$ and f be a real-valued function.

(1). $f(x)$ is said to converge to L as x *approaches* a **from the right** if and only if f is defined on some open interval I with left endpoint a and for every $\varepsilon > 0$ there is a $\delta > 0$ such that

$$a + \delta \in I \text{ and } a < x < a + \delta \text{ imply } |f(x) - L| < \varepsilon. \tag{2}$$

In this case we call L the *right-hand limit* of f at a, and denote it by

$$f(a^+) := L =: \lim_{x \to a^+} f(x) \ .$$

(2). $f(x)$ is said to converge to L as x approaches a **from the left** if and only if f is defined on some open interval I with right endpoint a and for every $\varepsilon > 0$ there is a $\delta > 0$ such that $a - \delta \in I$ and $a - \delta < x < a$ imply $|f(x) - L| < \varepsilon$, In this case we call L the *left-hand limit* of f at a and denote it by

$$f(a^-) := L = \lim_{x \to a^-} f(x) \ .$$

Example 1

(i). Prove that

$$f(x) = \begin{cases} x+1 & x \geq 0 \\ x-1 & x < 0 \end{cases}$$

has one-sided limits at $a = 0$ but $\lim_{x\to 0} f(x)$ does not exist.

(ii). Prove that $\lim\limits_{x\to 0^+} \sqrt{x} = 0$.

Proof.

(i). Let $\varepsilon > 0$ and set $\delta = \varepsilon$. If $0 < x < \delta$, then $|f(x) - 1| = |x| < \delta = \varepsilon$. Hence $\lim\limits_{x\to 0^+} f(x)$ exists and equals 1. Similarly, $\lim\limits_{x\to 0^-} f(x)$ exists and equals -1. However, $x_n = (-1)^n/n \to 0$ but $f(x_n) = (-1)^n(1+1/n)$ does not converge as $n \to \infty$. Hence by the Sequential Characterization of limits, $\lim\limits_{x\to 0} f(x)$ does not exist.

(ii). Let $\varepsilon > 0$ and set $\delta = \varepsilon^2$. If $0 < x < \delta$, then $|f(x)| = \sqrt{x} < \sqrt{\delta} = \varepsilon$.

□

Remark : Not every function has one-sided limits (see **Example 4** in Note 1). The two parts in **Example 1** above show that even when a function has one-sided limits, it may not have a two-sided limit. The following result, however, shows that if both one-sided limits, at a point a, exist and are EQUAL, then the two-sided limit at a exists.

Theorem 4.2.1 : The limit $\lim\limits_{x\to a} f(x)$ exists and equals L if and only if

$$L = \lim_{x\to a^+} f(x) = \lim_{x\to a^-} f(x). \tag{3}$$

Proof.

"$\Longrightarrow$"

- If the limit L of $f(x)$ exists as $x \to a$, then given $\varepsilon > 0$ choose $\delta > 0$ such that $0 < |x - a| < \delta$ implies $|f(x) - L| < \varepsilon$.
- Since any x which satisfies $a < x < a + \delta$ or $a - \delta < x < a$ also satisfies $0 < |x - a| < \delta$, it is clear that both the left and right limits of $f(x)$ exist as $x \to a$ and satisfy (3).

"$\Longleftarrow$"

- Suppose that (3) holds. Then given $\varepsilon > 0$ there exists a $\delta_1 > 0$ (respectively, a $\delta_2 > 0$) such that $a < x < a + \delta_1$ (respectively, $a - \delta_2 < x < a$) implies $|f(x) - L| < \varepsilon$.

- Set $\delta = \min\{\delta_1, \delta_2\}$. Then $0 < |x - a| < \delta$ implies $a < x < a + \delta_1$ or $a - \delta_2 < x < a$ (depending on whether x is to the right or to the left of a).

- Hence (1) holds; that is, $f(x) \to L$ as $x \to a$.

□

The definition of limits can be expanded to include extended real numbers.

Definition 4.2.2 : Let $a, L \in \mathbb{R}$.

(i). $f(x) \to L$ as $x \to \infty$ if and only if there exists a $c > 0$ such that $(c, \infty) \subset \text{Dom}\ (f)$ and given $\varepsilon > 0$ there is an $M \in \mathbb{R}$ such that $x > M$ implies $|f(x) - L| < \varepsilon$, in which case we shall write

$$\lim_{x \to \infty} f(x) = L \quad \text{or} \quad f(x) \to L \text{ as } x \to \infty\ .$$

Similarly, $f(x) \to L$ as $x \to -\infty$ if and only if there exists a $c > 0$ such that $(-\infty, -c) \subset \text{Dom}\ (f)$ and given $\varepsilon > 0$ there is an $M \in \mathbb{R}$ such that $x < M$ implies $|f(x) - L| < \varepsilon$, in which case we shall write

$$\lim_{x \to -\infty} f(x) = L \quad \text{or} \quad f(x) \to L \text{ as } x \to -\infty\ .$$

(ii). $f(x) \to \infty$ as $x \to a$ if and only if there is an open interval I containing a such that $I \backslash \{a\} \subset \text{Dom}(f)$ and given $M \in \mathbb{R}$ there is a $\delta > 0$ such that $0 < |x - a| < \delta$ implies $f(x) > M$, in which case we shall write

$$\lim_{x \to a} f(x) = \infty \quad \text{or} \quad f(x) \to \infty \text{ as } x \to a\ .$$

Similarly, $f(x) \to -\infty$ as $x \to a$ if and only if there is an open interval I containing a such that $I \backslash \{a\} \subset \text{Dom}\ (f)$ and given $M \in \mathbb{R}$ there is a $\delta > 0$ such that $0 < |x - a| < \delta$ implies $f(x) < M$, in which case we shall write

$$\lim_{x \to a} f(x) = -\infty \quad \text{or} \quad f(x) \to -\infty \text{ as } x \to a\ .$$

Example 2

(i). Prove that $1/x \to 0$ as $x \to \infty$.

(ii). Prove that $\lim_{x\to 1^-} f(x) := \lim_{x\to 1^-} \dfrac{x+2}{2x^2-3x+1} = -\infty$.

Proof.

(i).

- Given $\varepsilon > 0$, set $M = 1/\varepsilon$.
- If $x > M$, then $|1/x| = 1/x < 1/M = \varepsilon$.
- Thus $1/x \to 0$ as $x \to \infty$.

(ii).

- Let $M \in \mathbb{R}$. We must show that $f(x) < M$ for x near but to the left of 1 (no matter how large and negative M is).
- Without loss of generality, assume that $M < 0$.
- As $x \to 1^-$, $2x^2 - 3x + 1$ is negative and converges to 0. (Observe that $2x^2 - 3x + 1$ is a parabola opening upward with roots $1/2$ and 1.)
- Therefore, choose $\delta \in (0, 1)$ such that $1 - \delta < x < 1$ implies $\dfrac{2}{M} < 2x^2 - 3x + 1 < 0$; that is, $\dfrac{-1}{(2x^2-3x+1)} > \dfrac{-M}{2} > 0$.
- Since $0 \le x < 1$ also implies $2 < x + 2 < 3$, it follows that $-(x+2)/(2x^2-3x+1) > -M$ that is,

$$f(x) = \frac{x+2}{2x^2-3x+1} < M \quad \text{for all} \quad 1-\delta < x < 1$$

□

In order to unify the presentation of one-sided, two-sided, and infinite limits, we introduce the following notation. Let a be an extended real number, and let I be a non-degenerate open interval that either contains a or has a as one of its endpoints, Suppose further that f is a function defined on I except possibly at a. If a is finite and I contains a, then

$$\lim_{\substack{x\to a \\ x\in I}} f(x) \tag{4}$$

will denote $\lim_{x\to a} f(x)$ (when it exists); if a is a finite left endpoint of I, then (4) will denote $\lim_{x\to a^+} f(x)$ (when it exists); if a is a finite right endpoint of I, then (4) will denote $\lim_{x\to a^-} f(x)$ (when it exists); if $a = \pm\infty$ is an endpoint of I, then (4) will denote $\lim_{x\to\pm\infty} f(x)$ (when each exists).

Using this notation, we can state a **Sequential Characterization of limits** valid for two-sided, one-sided, and infinite limits.

Theorem 4.2.2 : Let a be an extended real number, and let I be a non-degenerate open interval that either contains a or has a as one of its endpoints. Suppose further that f is a function defined on I except possibly at a. Then

$$\lim_{\substack{x\to a \\ x\in I}} f(x)$$

exists and equals L if and only if $f(x_n) \to L$ for all sequences $\{x_n\} \in I$ which satisfy $x_n \neq a$ and $x_n \to a$ as $n \to \infty$.

Proof. Since we have already proved this for finite two-sided limits, we must show it for the remaining eight cases which notation (4) represents. Since the proofs are similar, we shall give the details for only one of these cases, namely the case when a belongs to I and $L = \infty$. Thus we must prove that $f(x) \to \infty$ as $x \to a$ if and only if $f(x_n) \to \infty$ for any sequence $\{x_n\} \in I$ which converges to a and satisfies $x_n \neq a$ for $n \in \mathbb{Z}^+$.

"$\Longrightarrow$" :

- Suppose first that $f(x) \to \infty$ as $x \to a$.
- If $\{x_n\} \in I$, $x_n \to a$ as $n \to \infty$, and $x_n = a$, then given $M \in \mathbb{R}$ there is a $\delta > 0$ such that $0 < |x-a| < \delta$ implies $f(x) > M$, and there is an $N \in \mathbb{Z}^+$ such that $n \geq N$ implies $|x_n - a| < \delta$.
- Consequently, $n \geq N$ implies $f(x_n) > M$; that is, $f(x_n) \to \infty$ as $n \to \infty$ as required.

"$\Longleftarrow$" :

- Suppose to the contrary that $f(x_n) \to \infty$ for any sequence $\{x_n\} \in I$ which converges to a and satisfies $x_n = a$ but $f(x)$ does NOT converge to ∞ as $x \to a$.

- By the definition of "convergence" to ∞ there are numbers $M_0 \in \mathbb{R}$ and $x_n \in I$ such that $|x_n - a| < 1/n$ and $f(x_n) \leq M_0$ for all $n \in \mathbb{Z}^+$.

- The first condition implies $x_n \to a$ but the second condition implies that $f(x_n)$ does not converge to ∞ as $n \to \infty$.

- This contradiction proves **Theorem 4.2.2** in the case $a \in I$ and $L = \infty$.

□

Using **Theorem 4.2.2** we can prove limit theorems that are function analogues of **Theorem 3.2.4** and its **Corollary** (See **Section 3.2 : Limit Theorems**).

These limit theorems can be used to evaluate infinite limits and limits at $\pm\infty$.

Example 3 Prove that $\lim_{x\to\infty} \dfrac{2x^2 - 1}{1 - x^2} = -2$.

Proof.

- Since the limit of a product is the product of the limits, we have by **Example 2 (i)** that $1/x^m \to 0$ as $x \to \infty$ for any $m \in \mathbb{Z}^+$.

- Multiplying the numerator and the denominator of the expression above by $1/x^2$, we obtain

$$\lim_{x\to\infty} \frac{2x^2 - 1}{1 - x^2} = \lim_{x\to\infty} \frac{2 - 1/x^2}{-1 + 1/x^2} = \frac{\lim_{x\to\infty}(2 - 1/x^2)}{\lim_{x\to\infty}(-1 + 1/x^2)} = \frac{2}{-1} = -2 \ .$$

□

.....

Exercises

(1). Decide which of the following statements are true and which are false. Prove the true ones and provide counterexamples for the false ones.

(a). If $f(x) \to \infty$ as $x \to \infty$ and $g(x) > 0$, then $g(x)/f(x) \to 0$ as $x \to \infty$.

(b). If $f(x) \to 0$ as $x \to a^+$ and $g(x) \geq 1$ for all $x \in \mathbb{R}$, then $g(x)/f(x) \to \infty$ as $x \to a^+$.

(c). If $f(x) \to \infty$ as $x \to \infty$, then $\sin\left(x^2 + x + 1\right)/f(x) \to 0$ as $x \to \infty$.

(d). If P and Q are polynomials such that the degree of P is less than or equal to the degree of Q (see **Exercise 5**), then there is an $L \in \mathbb{R}$ such that

$$\lim_{x\to\infty} \frac{P(x)}{Q(x)} = \lim_{x\to-\infty} \frac{P(x)}{Q(x)} = L.$$

(2). For each of the following, use definitions (rather than limit theorems) to prove that the limit exists. Identify the limit in each case.

(a). $\displaystyle\lim_{x\to0^-} \frac{\sqrt{x^2}}{x}$

(b). $\displaystyle\lim_{x\to\infty} \frac{\sin x}{x^2}$

(c). $\displaystyle\lim_{x\to-1^+} \frac{1}{x^2-1}$

(d). $\displaystyle\lim_{x\to1^+} \frac{x-3}{3-x-2x^2}$

(e). $\displaystyle\lim_{x\to-\infty} \frac{\cos(\tan x)}{x+1}$

(3). Assuming that $e^x \to e^a, \sin x \to \sin a$, and $\cos x \to \cos a$ as $x \to a$ for any $a \in \mathbb{R}$, evaluate the following limits when they exist.

(a). $\displaystyle\lim_{x\to2^-} \frac{x^3-x^2-4}{x^2-4}$

(b). $\displaystyle\lim_{x\to\infty} \frac{5x^2+3x-2}{3x^2-2x+1}$

(c). $\displaystyle\lim_{x\to-\infty} e^{-1/x^2}$

(4). Prove the following comparison theorems for the functions f and g, and $a \in \mathbb{R}$.

(a). If $f(x) \geq g(x)$ and $g(x) \to \infty$ as $x \to a$, then $f(x) \to \infty$ as $x \to a$.

(b). If $f(x) \leq g(x) \leq h(x)$ and

$$L := \lim_{x \to \infty} f(x) = \lim_{x \to \infty} h(x)$$

then $g(x) \to L$ as $x \to \infty$.

(5). Recall that a polynomial of degree n is a function of the form

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 \ ,$$

where $a_j \in \mathbb{R}$ for $j = 0, 1, \ldots, n$ and $a_n \neq 0$.

(a). Prove that if $0^0 = 1$, then $\lim_{x \to a} x^n = a^n$ for $n = 0, 1, \cdots$ and $a \in \mathbb{R}$.

(b). Prove that if P is a polynomial, then

$$\lim_{x \to a} = P(a)$$

for every $a \in \mathbb{R}$.